PRODUCT BUILD • MVP OVERVIEW

AI-assisted visual storytelling for graphic novels, illustrated stories, and film storyboards. This report summarizes the MVP's tools, architecture, and current deployment scope.

01 Project objective

Vizzy Studio helps a creator move from a story idea to an ordered sequence of illustrated scenes. The MVP combines a guided project brief, reusable visual style and character context, outline review, image alternatives, version selection, page approval, and a slideshow for the finished sequence.

02 What was used

AREA	TECHNOLOGY / ROLE
Web application	**React 19 + TypeScript** for the studio interface; **Vite** for local development and production builds.
UI & routing	**Tailwind CSS 4** and custom CSS; **React Router** for navigation; **TanStack Query** for server data; **Zustand** for lightweight client state.
API	**Node.js + Express 5 + TypeScript** for authenticated REST endpoints, validation, media delivery, and generation-job processing.
Persistence	**MongoDB + Mongoose** models for users, projects, pages, characters, chat, image assets, and generation jobs.
AI & images	**OpenAI API** integration for story planning/chat and image generation/editing; configurable **ComfyUI** workflow as a self-hosted open-source image option.
Storage & tests	**Local disk** in development or optional **Cloudinary**; Vitest, Supertest, and Testing Library for automated checks.

03 MVP architecture at a glance

CREATOR
React studio

Brief, style guide, characters, page review, slideshow

→

EXPRESS API
auth · projects · jobs

Zod validation, ownership checks, queued generation

→

DATA & MEDIA
MongoDB · image storage

Persistent assets, alternatives, versions, page order

FEATURES • SAFETY • DEPLOYMENT

How the MVP works

1	**Set up a story**	Create a project brief with genre, audience, page goal, aspect ratio, art style, palette, lighting, and creative notes.
2	**Build continuity**	Create reusable character profiles and attach visual reference images; keep a shared style guide for future scenes.
3	**Plan and approve**	Ask the AI to draft a structured outline, review it, and explicitly approve scenes before they become draft pages.
4	**Generate and refine**	Submit image jobs for alternatives, poll job status, choose a version, or refine from a selected image without overwriting earlier versions.
5	**Finish and present**	Edit page text, reorder pages, approve completed pages, and play approved pages in a timed, looping slideshow; export an approved-page HTML view.

04 Security and reliability included

AUTHENTICATION	DATA & INPUTS	GENERATION
bcrypt password hashing; signed JWT in an HTTP-only cookie; per-project ownership checks.	Zod schemas; Helmet; restricted credentialed CORS; API/auth rate limits; upload size/type checks; centralized safe errors.	Persistent, idempotent job requests; retry support; alternatives and image versions; generation limits; recoverable work queue.

05 Verification and deployment

Automated checks

Production client build and TypeScript check passed. ESLint passed. Vitest passed: **6 tests** across API-boundary, prompt-construction, and onboarding UI coverage. Live MongoDB/OpenAI/Cloudinary/ComfyUI calls were not tested without service configuration.

Public share link

shobhithub.github.io/Project-Storyboard-Creator
Deployed with GitHub Actions / GitHub Pages.

IMPORTANT SCOPE NOTE

The public link is the **browser-only demo**: it uses localStorage and demo artwork, so each visitor's work stays in that browser. The MongoDB-backed API and real AI providers are implemented in the project but are **not connected to this Pages deployment**. A full multi-user production launch still requires a hosted API, MongoDB, a private JWT secret, and provider/storage credentials configured in hosting secrets.

Prepared 18 October 2026 • Project repository: github.com/Shobhithub/Project-Storyboard-Creator